

Flight periods and life history strategies of caddisflies in temporary and permanent woodland brooks in the Lower Rhine area (Germany)

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Abstract. A total of 71 species of caddisflies was collected during a 3-year-study by light traps and larval samples from 52 mostly unpolluted brooks in the Lower Rhine area (30-70 m a.s.l.). Almost half of the streams run dry in summer for several months because of the specific geological and hydrological conditions. Caddisflies of these temporary brooks show several adaptations associated with colonization of such habitats which are reflected, for example, in the flight periods: a short flight period in early spring followed by a larval quiescence in summer or an extended flight period from spring into autumn with diapausing adults are typical for 73% of species which occur primarily in temporary brooks. Permanent streams are colonized by species which have their flight periods in summer or in autumn without diapausing adults (96%). Autecological studies of the rare phryganeid species *Oligostomis reticulata* L. in temporary brooks showed a new type of life cycle with a long larval quiescence in the dry stream bed.

The West European lowland areas are heavily impacted by human activities. Most of the streams are regulated to drain water basins and their former forested environment has been degraded for intensive land use. However, on the sand flats of the Lower Rhine area (Fig. 1), there is still a considerable number of nearly natural streams due to the special geological, geographic, and historical conditions.

Since 1991 hydrological, chemical, and biological investigations have been conducted on 52 brooks, about half of them being temporary because of a labile groundwater table. Caddisflies have been selected as a well known group of aquatic insects to investigate specific adaptations enabling the colonization of the different types of brooks.

Study Area

The sand flats of the Lower Rhine (30-70 m a.s.l.) are situated on the eastern side of the Rhine, between the northwestern margin of the conurbation Ruhrgebiet and the Dutch border (total area about 570 km²). A layer of poor glacial and fluvial sands with gravels covering a thick stratum of clay preserved the landscape from intensive agricultural land use and settlement. Therefore nearly half of the sand flats are still covered by forests, most of the streams follow their natural bed and are accompanied by alder (*Alnus glutinosa*). The hydrological conditions

in combination with the evapotranspiration are responsible for the drying out of numerous streams during summer (June to October).

In this area, two basic types of streams can be distinguished, the "organic type" and the "mineral type" (Sommerhäuser and Timm 1993, Timm and Sommerhäuser 1993). The substrates of the slow-flowing organic-type streams are characterized by a

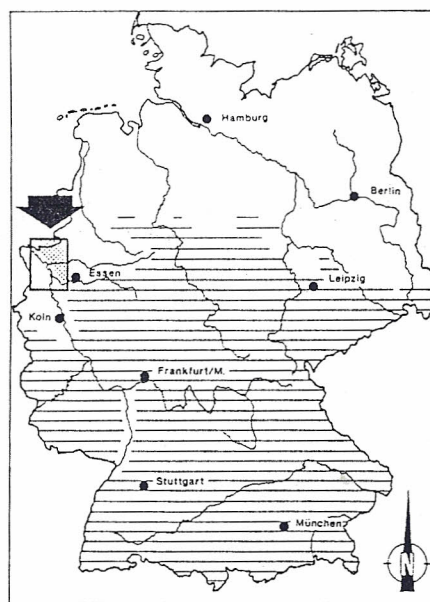


FIG. 1. Map of Germany showing the location of the study area as part of the North German lowlands, near the Dutch border. White areas = lowland regions (< 100 m a.s.l.), shaded areas = mountainous regions (> 100 m a.s.l.), dotted area = study area.

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layer of coarse and fine detritus, which may be quite thick, throughout the year. On the banks, the water table is less than 0.5 m below ground level, the water is acid (pH 4 to 6) and soft (4 to 8° on the German scale). Flow velocities range between <0.1 to 0.5 m/s, the morphology of the channel can be described as anastomosing in a broad flat valley. The vegetation is characterized by soft water plants like *Potamogeton polygonifolius*, *Scapania undulata* and *Sphagnum* spp. In this type of brook the phases of drying out and refilling of the stream bed at the beginning and end of the dry period last several days to some weeks. The regime of discharge is moderated by the thick organic layer (sometimes peat).

Mineral-type streams are characterized by mineral substrates (sand, gravels) throughout the year and are meandering more than 0.5 m beneath their surroundings. The water table is 0.5 to 1 m below ground on the banks and flow velocities range from 0.2 to 0.6 m/s. The water is approximately neutral (pH 6.5 to 7.5), with a total hardness of 7 to 13° on the German scale. Characteristic water plants are *Fontinalis antipyretica* and *Glyceria fluitans*. The phases of drying out and refilling of the stream bed at the beginning and end of the dry phase are rather short (several hours to a few days).

Methods

Measuring of flow and analysis of oxygen saturation and ammonium (NH_4^+) were conducted in monthly intervals for about one year to describe abiotic conditions of temporary and permanent brooks.

Adult Trichoptera were sampled over two flight periods (1992, 1993; April-October) at nine selected brooks representing the different types of streams in the area investigated. Two light traps each with three lamps (UV-visible, black and working light) in a gauze tower (height 2 m, diameter 0.9 m) were placed near the stream valley at 2 to 4 week intervals. Specimens attracted by light were collected by hand. Caddis larvae were sampled in 52 brooks by hand net or Surber sampler over three years.

Autecological studies were conducted on *Oligostomis reticulata* L. (Phryganeidae). Larvae were sampled monthly in the Steinbach, a small first order stream (according to Strahler 1957) which dries up during summer. Laboratory experiments were conducted to analyse larval ability to survive drying up of the habitat and the type of dormancy.

Results

Hydrological and physico-chemical conditions in temporary streams

Figure 2 shows changes of abiotic conditions in the Steinbach, an unpolluted temporary brook, over one year. During the period of flow (Oct-May) discharge normally ranges between 3 to 12 l/s, ammonium does not exceed 0.3 mg/l, the water is nearly oxygen saturated (80-100%) and quite cold (0-15°C). Between the end of May and October the flow stops. Measurements in summer months in remaining pools show rapidly increasing ammonium contents (exceeding 1.4 mg/l) in combination with decreasing oxygen saturation (below 20%). Temperature in the shady pools of the woodland streams, however, usually never exceeds 20°C.

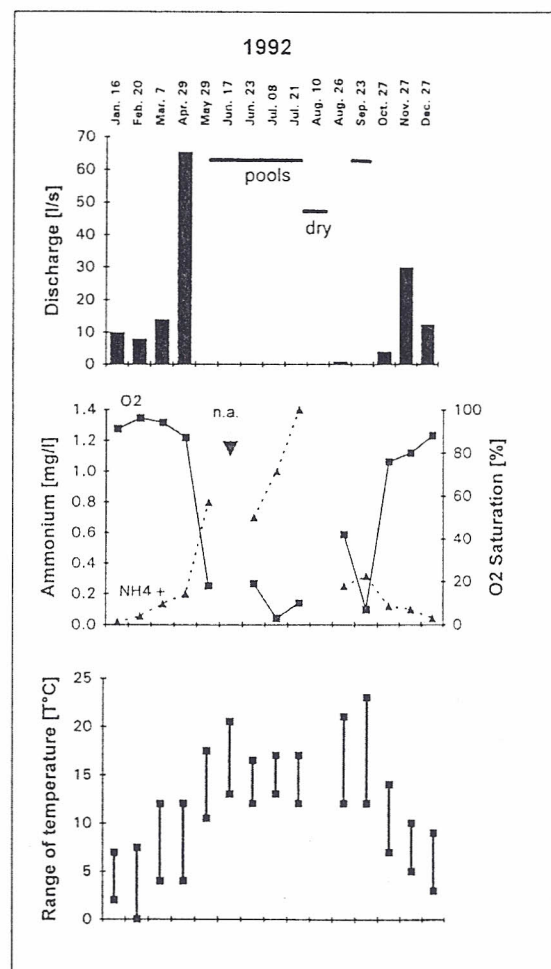


FIG. 2. Discharge, oxygen saturation, ammonium content and water temperature in an unpolluted temporary brook over one year. Ranges of water temperature were measured by minimum-maximum-thermometer, each bar represents maximum and minimum values of the previous measuring period. n.a. = not analyzed.

Species distribution in stream types

A total of 71 species was determined, belonging to 15 families (Table 1). Streams of the organic type can be characterized by the absence of species belonging to the families Rhyacophilidae, Glossosomatidae, Hydropsychidae, and Psychomyiidae,

nearly all Leptoceridae and the limnephilid genera *Potamophylax*, *Halesus*, and *Anabolia*, most of them being abundant in the different brooks of the mineral type. The typical group of caddisflies in the organic-type brook contains the polycentropodid *Plectrocnemia conspersa* Curtis, the limnephilids *Micropterna lateralis* Stephens, *Glyphotaelius*

TABLE 1. Species list of caddisflies caught by light trap or hand net in the streams of the Lower Rhine sand flats during 1991-1993, with remarks on the stream type inhabited (organic type brook/mineral type brook; temporary/permanent stream type). 1 = stream type uncertain (no larvae caught); 2 = larvae terrestrial; + = more frequent; (+) = rarely; - = never caught; and, ? = expected.

	Stream type and number (N) examined			
	Organic (17)	Mineral (28)	Temporary (19)	Permanent (26)
RHYACOPHILIDAE				
<i>Rhyacophila fasciata</i> Hagen	-	(+)	-	(+)
GLOSSOSOMATIDAE				
<i>Agapetus fuscipes</i> Curtis	-	(+)	-	(+)
HYDROPTILIDAE ¹				
<i>Orthotrichia costalis</i> Curtis				
<i>Oxyethira flavicornis</i> Pictet				
<i>Hydroptila cf. angulata</i> Mosely				
<i>Hydroptila tineoides</i> Dalman				
<i>Agraylea sexmaculata</i> Curtis				
HYDROPSYCHIDAE				
<i>Hydropsyche angustipennis</i> Curtis	-	+	-	+
<i>Hydropsyche contubernalis</i> McL.	-	(+)	-	(+)
<i>Hydropsyche pellucidula</i> Curtis	-	+	-	+
<i>Hydropsyche saxonica</i> McL.	-	+	-	+
<i>Hydropsyche siltalai</i> Döhler	-	+	-	+
POLYCENTROPODIDAE				
<i>Neureclipsis bimaculata</i> L.	-	(+)	-	(+)
<i>Plectrocnemia conspersa</i> Curtis	+	+	+	+
<i>Polycentropus irroratus</i> Curtis	-	+	-	+
<i>Cyrnus flavidus</i> McL.	-	+	-	+
<i>Cyrnus trimaculatus</i> Curtis	(+)	+	(+)	+
PSYCHOMYIIDAE				
<i>Lype phaeopa</i> Stephens	?	(+)	-	(+)
<i>Lype reducta</i> Hagen	-	+	-	+
<i>Tinodes assimilis</i> McL.	-	+	-	+
<i>Tinodes pallidulus</i> McL.	-	(+)	-	(+)
<i>Tinodes waeneri</i> L.	-	+	-	+
ECNOMIDAE ¹				
<i>Ecnomus tenellus</i> Ramb.				
PHRYGANEIDAE				
<i>Trichostegia minor</i> Curtis	+	-	+	+
<i>Agrypnia varia</i> Fabr.	+	+	+	+
<i>Phryganea grandis</i> L.	-	+	+	+
<i>Oligotricha striata</i> L.	(+)	(+)	(+)	(+)
<i>Oligostomis reticulata</i> L.	+	+	+	-
LIMNEPHILIDAE				
<i>Ironoquia dubia</i> Stephens	-	(+)	(+)	-

TABLE 1. Species list of caddisflies, continued.

	Stream type and number (N) examined			
	Organic (17)	Mineral (28)	Temporary (19)	Permanent (26)
LIMNEPHILIDAE, CONTINUED.				
<i>Limnephilus auricula</i> Curtis	+	+	+	+
<i>Limnephilus extricatus</i> McL.	-	+	-	+
<i>Limnephilus flavicornis</i> Fabr.	-	+	-	+
<i>Limnephilus lunatus</i> Curtis	-	+	+	+
<i>Limnephilus rhombicus</i> L.	-	+	+	+
<i>Limnephilus sparsus</i> Curtis	+	+	+	+
<i>Grammotaulius submaculatus</i> Ramb.	(+)	-	(+)	?
<i>Glyptotaelius pellucidus</i> Retz.	+	+	+	+
<i>Anabolia nervosa</i> Curtis	-	+	-	+
<i>Potamophylax cingulatus</i> Steph.	-	+	-	+
<i>Potamophylax luctuosus</i> Pill. & Mitt.	-	+	-	+
<i>Potamophylax cingulatus/luctuosus</i> Gr.	-	+	-	+
<i>Potamophylax rotundipennis</i> Brauer	-	+	-	+
<i>Halesus digitatus</i> Schrank	-	+	-	+
<i>Halesus radiatus</i> Curtis	(+)	+	-	+
<i>Enoicyla pusilla</i> Burm. ²				
<i>Stenophylax permistus</i> McL.	+	+	+	+
<i>Micropterna lateralis</i> Steph.	+	(+)	+	(+)
<i>Micropterna sequax</i> McL.	(+)	+	+	+
GOERIDAE				
<i>Goera pilosa</i> Fabr.	-	+	-	+
<i>Lithax obscurus</i> Hagen	-	(+)	(+)	?
LEPIDOSTOMATIDAE				
<i>Lasiocephala basalis</i> Kol.	-	(+)	-	(+)
LEPTOCERIDAE				
<i>Athripsodes aterrimus</i> Steph.	-	+	-	+
<i>Athripsodes bilineatus</i> L.	-	+	-	+
<i>Athripsodes cinereus</i> Curtis	-	+	-	+
<i>Ceraclea alboguttata</i> Hagen ¹				
<i>Ceraclea dissimilis</i> Steph ¹				
<i>Mystacides azurea</i> L.	-	+	-	+
<i>Mystacides longicornis</i> L.	-	+	-	+
<i>Mystacides nigra</i> L.	-	+	-	+
<i>Mystacides</i> sp.	-	+	-	+
<i>Oecetis lacustris</i> Pictet ¹				
<i>Oecetis notata</i> Ramb. ¹				
<i>Oecetis ochracea</i> Curtis ¹				
<i>Adicella reducta</i> McL.	+	+	?	+
SERICOSTOMATIDAE				
<i>Notidobia ciliaris</i> L.	-	+	-	+
<i>Sericostoma personatum</i> K. & Sp.	?	+	-	+
BERAEIDAE				
<i>Beraea maura</i> Curtis	(+)	?	(+)	?
<i>Beraea pullata</i> Curtis	-	+	+	+
<i>Beraeodes minutus</i> L.	-	+	(+)	+
MOLANNIDAE				
<i>Molanna angustata</i> Curtis	-	+	-	+

pellucidus Retzius and *Limnephilus sparsus* Curtis, and sometimes accompanied by the phryganeids *Oligostomis reticulata* L. and *Trichostegia minor* Curtis (Phryganeidae).

Flight periods

Adults of 62 species were caught by light traps and hand net. Figure 3 shows flight periods of 56

species, differentiated for species occurring only or mainly in temporary streams (3a) and species which are restricted to permanent streams (3b). Long flight periods extending to six months are characteristic for caddisflies in temporary brooks while those of permanent brooks mainly have flight periods shorter than two months (Fig. 4). The presence of species in each month of the whole flying period is different between temporary and permanent brooks (Fig. 5): the first

Species / Part of month	Type of flight period	APR		MAI		JUN		JUL		AUG		SEP		OCT		Duration of flight period (acc. lit. *)
		I	II	I	II	I	II	I	II	I	II	I	II	I	II	
<i>Oligostomis reticulata</i>	A															III-V
<i>Lithax obscurus</i>	A															V-VI
<i>Micropterna lateralis</i>	B															V-X
<i>Stenophylax permistus</i>	C															V-X
<i>Plectrocnemia conspersa</i>	B															VI-X
<i>Limnephilus sparsus</i>	C															V-X
<i>Glyptotaelius pellucidus</i>	C															V-X
<i>Beraea maura</i>	B															V-VIII
<i>Limnephilus rhombicus</i>	C															V-X
<i>Micropterna sequax</i>	C															V-X
<i>Limnephilus auricula</i>	C															V-X
<i>Limnephilus extricatus</i>	C															V-X
<i>Trichostegia minor</i>	B															IV-VIII
<i>Limnephilus lunatus</i>	C															V-XII
<i>Grammotaulius submaculatus</i>	B															VI-VII
<i>Limnephilus flavicornis</i>	C															V-X

FIG. 3A. Duration of flight periods of 16 species of caddisflies, caught only or mainly in temporary streams. Types of flight periods (A-D) according to definitions in Table 2. * = according to Tobias and Tobias (1981); line = imagines caught; and, dotted line = no imagines caught, but a continued flight period is supposed.

species in temporary streams can be caught early in spring (April), nearly 50% of the species are present in May and in autumn still 30 to 40% of the species are to be found as adults. In contrast, most species of permanent streams appear only in the summer months.

Crichton (1971) presented a helpful system to classify the flight periods of caddisflies defining three groups of flight periods (compare Table 2). Regarding the flight period and life cycle of *Oligostomis reticulata* L. as described below it may be supplemented by a further group: caddisflies with a short flight period in spring without an imaginal diapause. Table 2 shows the four types of flight periods proposed (A-D) and distribution of temporary and permanent stream species within the types. Sixty-two and one half % of the caddisflies in temporary brooks

have an extended flight period from spring to autumn, normally with a diapause. Twelve and one half % have a short flight period in spring as described for *Oligostomis reticulata*, 25% are summer-flying without diapausing adults. In permanent streams most Trichoptera belong to the summer-flying type (85%) or have a short flight period in autumn (10%).

Autecological studies of *Oligostomis reticulata* L.

An obviously undescribed type of flight period and survival strategy is represented by the phryganeid *Oligostomis reticulata*. *Oligostomis reticulata* nowadays is a rare trichopteran species which inhabits slow flowing dystrophic and marshy waters in Eurasia and Fennoscandia - habitats which in general have been affected by human activities. The life cycle of

Species / Part of month	Type of flight period	APR		MAY		JUN		JUL		AUG		SEP		OCT		Duration of flight period (acc. lit. *)
		I	II	I	II	I	II	I	II	I	II	I	II	I	II	
<i>Hydropsyche pellucidula</i>	B															V-X
<i>Notidobia ciliaris</i>	A															V-VI
<i>Beraeodes minutus</i>	A															V-VI
<i>Hydropsyche saxonica</i>	B															V-VI
<i>Tinodes assimilis</i>	B															V-IX
<i>Tinodes waeneri</i>	B															V-IX
<i>Cyrnus trimaculatus</i>	B															V-IX
<i>Ceraclea dissimilis</i>	B															VI-IX
<i>Mystacides azurea</i>	B															VI-X
<i>Tinodes pallidulus</i>	B															VII-IX
<i>Potamophylax luctuosus</i>	B															V-VII
<i>Lasiocephala basalis</i>	B															VI-IX
<i>Lype reducta</i>	B															V-IX
<i>Orthotrichia costalis</i>	B															V-VIII
<i>Sericostoma personatum</i>	B															VI-VIII
<i>Mystacides longicornis</i>	B															VI-IX
<i>Hydropsyche angustipennis</i>	B															IV-IX
<i>Hydropsyche siltalai</i>	B															VI-IX
<i>Polycentropus irroratus</i>	B															V-VII
<i>Ecnomus tenellus</i>	B															VI-IX
<i>Goera pilosa</i>	B															V-IX
<i>Oecetis lacustris</i>	B															VI-IX
<i>Oecetis notata</i>	B															VII-IX
<i>Oecetis ochracea</i>	B															VI-IX
<i>Hydropsyche contubernalis</i>	B															IV-X
<i>Hydroptila tineoides</i>	B															IV-IX
<i>Phryganea grandis</i>	B															IV-VIII
<i>Adicella reducta</i>	B															V-VIII
<i>Athripsodes cinereus</i>	B															V-IX
<i>Mystacides nigra</i>	B															V-IX
<i>Athripsodes aterrimus</i>	B															V-IX
<i>Molanna angustata</i>	B															V-IX
<i>Lype phaeopa</i>	B															IV-IX
<i>Hydroptila cf. angulata</i>	B															VI-VIII
<i>Oxyethira flavicornis</i>	B															VII-IX
<i>Ceraclea alboguttata</i>	B															VI-IX
<i>Cyrnus flavidus</i>	B															V-VII
<i>Agrypnia varia</i>	B															IV-IX
<i>Athripsodes bilineatus</i>	B															VI-VIII
<i>Potamophylax rotundipennis</i>	B															VIII-IX
<i>Anabolia nervosa</i>	D															VIII-XI
<i>Halesus digitatus</i>	D															VIII-XI
<i>Halesus radiatus</i>	D															IX-XI
<i>Potamophylax cingulatus</i>	D															VII-X **

FIG. 3B. Duration of flight periods of 44 species of caddisflies caught only in permanent streams. Types of flight periods (A-D) according to definitions in Table 2. * = according to Tobias and Tobias (1981). ** = according to Burkhard (1983). See Fig. 3A for further explanations.

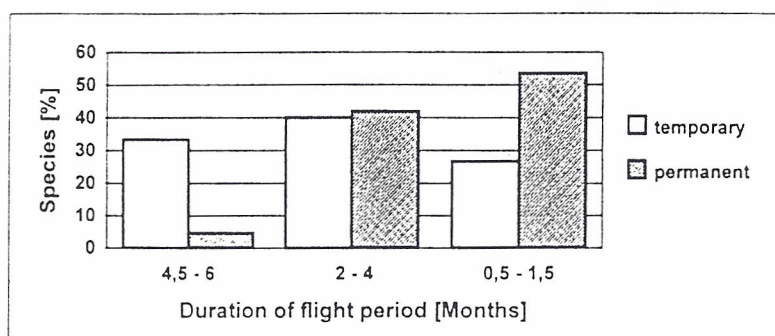


FIG. 4. Duration of flight periods of caddisfly species in the study area, differentiated for species of temporary and permanent streams.

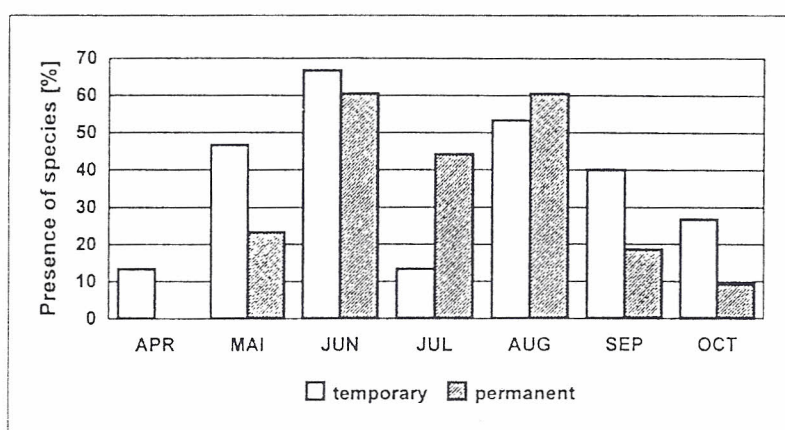


FIG. 5. Presence of species during the seven months of the flight period, differentiated for species of temporary and permanent streams

this phryganeid shows various adaptations which enable this species to be a successful colonizer of temporary streams. After a short flight period from the middle of April into May the first small larvae can be found abundant early in June (130 ind./m²).

With drying out of habitat at the end of June, larvae are immediately able to close their cases at both ends with a silken membrane which in combination with detritus particles seems to be an effective protection against desiccation during summer months. Larval

TABLE 2. Four types of flight periods (A-D) proposed by the authors and distribution of temporary and permanent stream species within the types, referring to Figs. 3A and 3B.

Type	Definition	Reference	% species	
			Temporary streams	Permanent streams
A	A short flight period in early spring, without a diapause.	authors	12.5	5.0
B	A shorter flight period, without a diapause, in spring and summer, sometimes extending into autumn.	Crichton (1971)	25.0	85.0
C	An extended flight period, probably involving a diapause, from spring through summer into autumn.	"	62.5	0
D	A short flight period, without a diapause, in autumn.	"	0	10.0

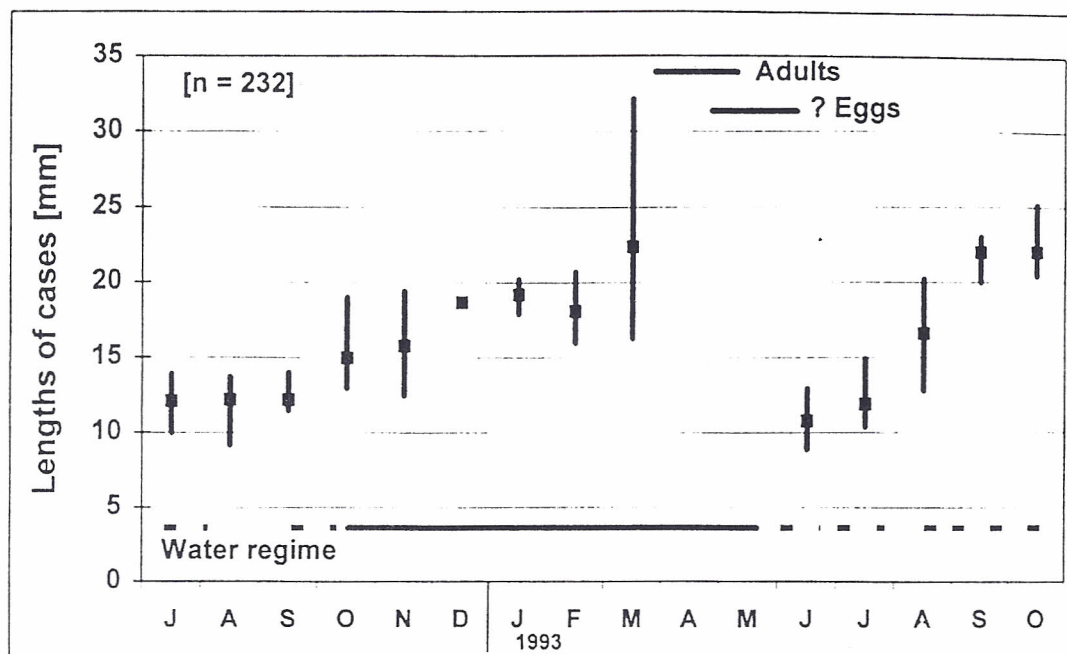


FIG. 6. Development of *Oligostomis reticulata* (Phryganeidae) during July 1992 to October 1993 in the Steinbach, a small first order stream. The figure shows lengths of cases (mean length, minimum and maximum), presence of adults and eggs. The line at the bottom represents water flow (line = period of flow, dotted line = period of isolated pools, no line = stream completely dry).

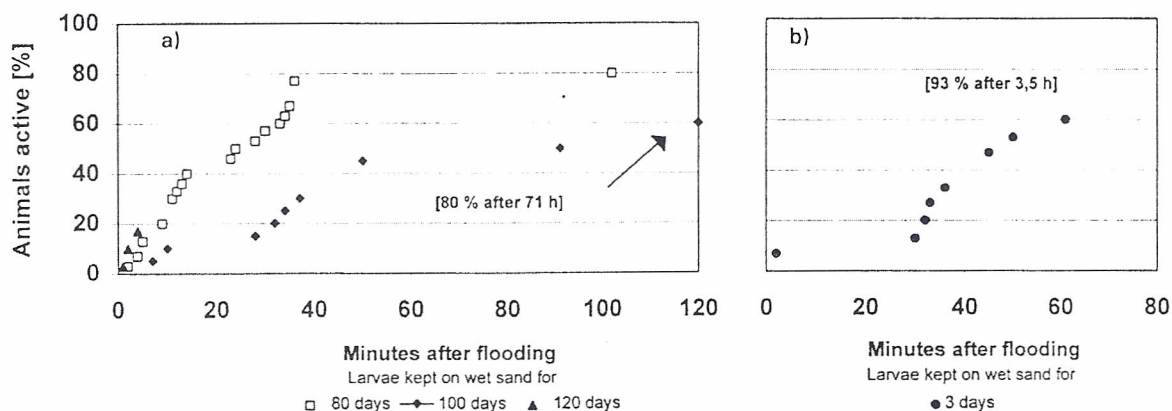


FIG. 7. Time required for *Oligostomis reticulata* larvae to reactivate after flooding experimental basins in which specimen have been kept on wet sand. A.—Larvae from 30.06.1993, kept over periods of 80, 100 and 120 days; B.—Larvae from 30.10.1993, kept over a period of 3 days.

development will be interrupted for up to four months (Fig. 6). With beginning of water flow in autumn larvae open the membrane and again begin to crawl and feed.

In laboratory experiments the closing and opening of the case can be easily induced, larvae reacting spontaneously on delivery and outlet of water in the experimental basins (Fig. 7). According to Müller (1970) this type of dormancy can be defined as quiescence. Furthermore, larvae of *O. reticulata* are able

to survive in remaining pools without a quiescence undergoing long periods of oxygen depletion and high ammonium contents.

Discussion

Successful colonizers of temporary streams show several adaptations which enable them to survive drying up in summer. Most caddisflies in temporary brooks of the Lower Rhine sand flats finish their lar-

val development before water flow rapidly decreases. They spend summer months as diapausing adults in the forested environment, egg laying takes place with the beginning of water flow in autumn (Stenophylacini, most Limnephilini). Other species are able to survive the dry season as young larvae in quiescence on the humid bottom of the brooks, sheltered against desiccation by closing their cases (e.g., *Oligostomis reticulata*). In both cases flight periods as integral parts of insect life cycles are synchronized with the hydrological regime.

The forested environment of the brooks examined is of importance for all life stages of caddisflies in summer giving humidity, shade, and shelter. The destruction of natural stream structures particularly threatens temporary waters. These ecological systems are as valuable elements in the hydrosphere as permanent waters, providing habitats for several highly specialized and sometimes rare species.

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